

Name: _____



Using the fx-991CW for statistics

Task A: Finding the mean from a frequency table

1) This frequency table shows the number of goals scored in each match by all the teams in a club over one football season.

Goals scored	Frequency
0	19
1	34
2	21
3	15
4	11

What was the mean number of goals scored?

2) This frequency table shows the star rating received by a restaurant over its opening weekend.

Star rating	Frequency
1	12
2	18
3	21
4	38
5	45

What was the mean star rating?

3) The frequency table shows the number of bedrooms in different houses within an area of a town.

Bedrooms	Frequency
1	20
2	93
3	54
4	33

What is the mean number of bedrooms?



4) The time it takes for a sample of pupils to travel to school is collected in this grouped frequency table.

Time taken to travel to school (in minutes)	Frequency (f)
$0 < t \leq 4$	28
$4 < t \leq 10$	23
$10 < t \leq 16$	70
$16 < t \leq 26$	52
$26 < t \leq 40$	30
$40 < t \leq 60$	13

What is the estimated mean travel time (to the nearest minute)?

Task B: Analysis

1) This frequency table shows the number of goals scored in each match by all the teams in a club over one football season.

Goals scored	Frequency
0	19
1	34
2	21
3	15
4	11

a) Calculate the median.

b) Calculate the interquartile range.

c) Would the club prefer to report the mean or median as the average number of goals scored? Why?

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Star rating	Frequency
1	12
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a) Calculate the median.

b) Calculate the interquartile range.

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3) The frequency table shows the number of bedrooms in different houses within an area of a town.

Bedrooms	Frequency
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A developer wishes to build more expensive houses (3+ bedrooms) but the local council want cheaper housing. The council argue that there are already enough houses with 3 or more bedrooms.

Are the council correct? Use the data to support your decision.

4) The time it takes for a sample of pupils to travel to school is collected in this grouped frequency table.

Time taken to travel to school (in minutes)	Frequency (f)
$0 < t \leq 4$	28
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It takes 17 mins for Jacob to travel to school by bus.

Jacob claims this takes him longer than the average pupil so he should be taken to school by car every day instead.

Do you agree or disagree with Jacob? Use evidence to support your answer.